# COMP3222 — Machine Learning Technologies Coursework

**Instructors:** Luis and Shoaib **Due Date:** 10 January 2025 at 4:00 PM.
**School:** ECS, University of Southampton
**Electronic Submission:** Online Submission
**Queries?** Please use the discussion forum for general questions. The forum is available on Blackboard. You are free to exchange ideas on the forum and ask questions from your fellow students. If you have something specific, e.g., an extenuating circumstance, please contact the ECS Student Office or the Relevant Board.

**Your "Individual" Experiments with a Clustering Algorithm**

## 1 Introduction

The $K$-means algorithm is a popular [1] clustering technique that "hard" partitions data into $K$ clusters. Figure 1 demonstrates the output from a converged $K$-means algorithm. However, determining the optimal number of clusters $K$ is a crucial yet challenging task. This coursework explores the application of Occam's Razor to guide the selection of the optimal $K$ value in $K$-means clustering.

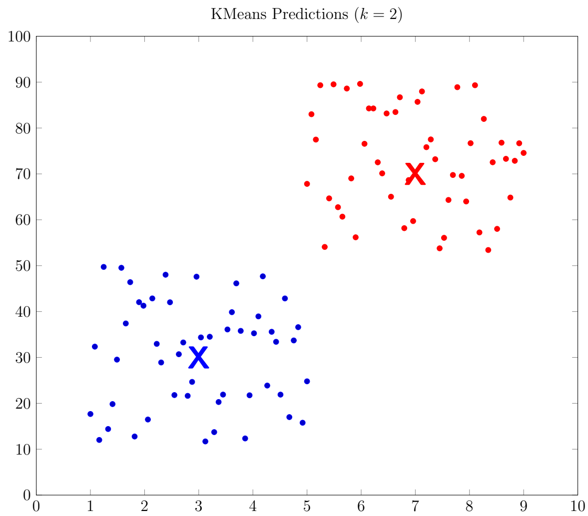


Figure 1: Illustrating the $K$-means model.

## 2 Key Learning Outcomes

We expect that independently doing this coursework could strengthen your foundations in machine learning and problem solving using data analysis. Through this coursework, we will establish the following learning outcomes:

1. Demonstrate knowledge of the $K$-means clustering algorithm and its underlying principles.
2. Use appropriate metrics to assess the quality of clustering results and compare different clustering solutions.
3. Understand the principle of Occam's Razor and apply it to select the optimal number of clusters in $K$-means.
4. Analyze and interpret the clusters obtained, gaining insights into the underlying patterns in the data.
5. Apply critical thinking and problem-solving skills to address clustering challenges and find optimal solutions.

You do not need to be a computer science expert to succeed in this coursework. While it will involve running some code, the focus is on understanding the concepts and applying them to real-world problems. The real challenge lies in using data-driven approaches to find meaningful solutions. This requires problem-solving skills and the ability to think critically about the data and the methods you are using. In-depth computer science programming knowledge is not as essential as your ability to work through challenges and achieve results.

## 3 Relevant Textbook Chapters

Relevant chapters from the textbook Machine Learning by Zhi-Hua Zhou

- Chapter 2
- Chapter 11

## 4 Coursework Problem

The problem that you must solve in this coursework is to find the optimal number of clusters $K$ in the $K$-means algorithm. This is a crucial task in clustering, as choosing an incorrect value of $K$ can lead to suboptimal results.

To this end, here is what you must do:

- Experiment with different values of $K$ and evaluate the clustering results using appropriate metrics.
- Consider the trade-off between model complexity (larger $K$ values) and model performance with Occam's Razor. Choose the simplest model that achieves satisfactory clustering results.
- Analyze the clusters obtained and gain insights into the underlying patterns in the data.

By effectively applying Occam's Razor, *in a principled way*, you can avoid overfitting and find the most appropriate number of clusters for their given dataset.

## 4.1 Executing the Problem

1. Download the text dataset [Download Link] that you will work with.

2. You may need to conduct appropriate processing of the dataset before you input the processed dataset into the $K$-means model. For instance, for datasets with missing values or outliers, preprocessing steps like imputation and normalization would be necessary.

3. You will then need to use *principled methods* in machine learning to find the optimal number of clusters.

4. Write a one-page report describing your solution.

## 4.2 Your One-Page PDF Report

An A4 page has two sides. By one page, we mean only the first/one side of your A4 document must contain your work, not the second page/side. For instance, when you open your document in a PDF reader, your PDF reader must only show 1 page of content. More than 1 page will incur a penalty. Another example, Page 1 of this coursework description contains the Introduction and Key Learning Outcomes.

Your one-page report must have:

- 11 point font size in Times New Roman A4 paper
- Contain your name and student ID
- Mention any preprocessing steps taken
- Explain how the K-means algorithm was used, what did you cluster?
- Describe the range of K values used
- Specify the metrics used to evaluate clustering performance. You must choose an appropriate metric used in clustering
- Explain how the principle of Occam's Razor was applied to select the optimal $K$ value
- Discuss the trade-off between model complexity and performance
- Present the clustering results for different K values
- Analyze the impact of Occam's Razor on the chosen K value

You could consider having the following in the one-page report:

- Tables - your own, not copied from external resources even when cited. Note that you just have one side of the A4 paper.
- Figures - you own, not copied from external resources even when cited. Note that you just have one side of the A4 paper.
- Only text description written by you
- References

# 5 What we are looking for in your coursework?

1. Using full one page. Following the font size mentioned above.
2. Citing relevant references. You may follow the referencing style which I have used in this document.
3. You mention your experimental settings followed by your results so that anyone can easily "replicate" your work. Missing information will result in a loss of marks. So, you simply have to summarise what you do to reach the ideal number of clusters. Copying and pasting commands in your report will not help, please refrain from writing commands, you must explain those commands if you want. Graphs, plots, and tables would be an excellent addition to the report.

# 6 Note on the marking criteria

Note on the marking criteria:

- Dataset processing (20%).
- $K$-means model selection and evaluation (70%). It is important to use principled methods in machine learning to address this. Lecture notes on model selection and evaluation have more details. [CAUTION:] Numerous online resources claim to automate the process of determining the number of clusters in $K$-means. However, our research indicates that many of these sources do not adhere to a principled machine learning paradigm for identifying cluster numbers. Simply replicating their solutions is unlikely to be beneficial.
- Report quality (10%). Is the report readable? Have you followed the guidelines, e.g., one-page A4 and font size?

# 7 Submission Materials

Submissions must consist of separate files. For example, if you have several source files, compress them into a single ZIP file. Therefore, you should submit two files: the ZIP file containing your source files and your PDF document. Consequently, each submission will include two files.

1. Your one-page PDF report.

2. Your source code. You may use any programming language and/or tool. As mentioned before, this is not a programming work, but a problem-solving problem. If you have used some tool to solve the problem, share the commands used to call and execute the tool. For instance, if you use MATLAB's $K$-means algorithm, share the MATLAB script, not the complete MATLAB software.

## 8 Need Help?

Please have a look at the following resources before attempting your coursework:

- Cross-validation tutorial Link 1
- Processing your dataset, e.g., text Link 2
- Parameter tuning Link 3
- Text clustering and the optimal number of topics/clusters Link 4
- Sample "unrelated" assignment Link 5

## 9 A Note of Plagiarism and Academic Honesty

Plagiarism is the act of presenting someone else's work as your own. This includes copying text, code, or ideas without proper attribution. Plagiarism is a serious academic offence and will be treated accordingly.

We will use different methods to detect copying from others and/or from the Internet. Some ways that we will employ are:

- Automated tools such as TurnItIn
- Manual comparison

*Note that will be extremely unlikely for two students to obtain almost similar conclusions and numerical results. This is because there is randomisation involved in the k-means algorithm. Besides that, the choice of data processing will significantly make the results different even on the same dataset.*

*It will also be very difficult to write the report without doing any experiments. We can easily detect such instances from the numerical results, code, and report.*

## References

[1] J. Xu, K. Xu, Y. Wang, Q. Shen, and R. Li. A k-means algorithm for financial market risk forecasting. *arXiv preprint arXiv:2405.13076*, 2024.